

Python Type Casting

```
In [1]: 110.34
```

```
Out[1]: 110.34
```

```
In [4]: int(110.34) # we can convert float to int
```

```
Out[4]: 110
```

```
In [5]: int(True) # boolean to int possible
```

```
Out[5]: 1
```

```
In [6]: int(1+2j)
```

```
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TypeError                                Traceback (most recent call last)  
Cell In[6], line 1  
----> 1 int(1+2j)  
  
TypeError: int() argument must be a string, a bytes-like object or a real number, not 'complex'
```

```
In [7]: int('10')
```

```
Out[7]: 10
```

```
In [8]: int('ten')
```

```
-----  
ValueError                                Traceback (most recent call last)  
Cell In[8], line 1  
----> 1 int('ten')  
  
ValueError: invalid literal for int() with base 10: 'ten'
```

in type casting while typecast from all other datatype to int possible except complex & text string

other datatype to float

```
In [9]: float(110)
```

```
Out[9]: 110.0
```

```
In [10]: float(True)
```

Out[10]: 1.0

In [11]: float(False)

Out[11]: 0.0

In [12]: float('10')

Out[12]: 10.0

In [13]: float('then')

```
-----  
ValueError                                Traceback (most recent call last)  
Cell In[13], line 1  
----> 1 float('then')  
  
ValueError: could not convert string to float: 'then'
```

other datatype to boolean

In [14]: bool(2)

Out[14]: True

In [15]: bool(0)

Out[15]: False

In [16]: bool(1+2j)

Out[16]: True

In [17]: bool('10')

Out[17]: True

In [18]: bool('then')

Out[18]: True

In [19]: bool()

Out[19]: False

In [20]: bool(-2)

Out[20]: True

all other datatype to complex

In [21]: complex(2)

Out[21]: (2+0j)

In [22]: `complex(2,4)`

Out[22]: (2+4j)

In [23]: `complex(3,4,6)`

```
-----  
TypeError                                Traceback (most recent call last)  
Cell In[23], line 1  
----> 1 complex(3,4,6)  
  
TypeError: complex() takes at most 2 arguments (3 given)
```

In [24]: `complex(2.3)`

Out[24]: (2.3+0j)

In [26]: `complex(2.4, 5.6)`

Out[26]: (2.4+5.6j)

In [29]: `complex(True)`

Out[29]: (1+0j)

In [30]: `complex(False)`

Out[30]: 0j

In [31]: `complex('10')`

Out[31]: (10+0j)

In [33]: `complex('then')`

```
-----  
ValueError                                Traceback (most recent call last)  
Cell In[33], line 1  
----> 1 complex('then')  
  
ValueError: complex() arg is a malformed string
```

In [35]: `complex('10', '20')`

```
-----  
TypeError                                Traceback (most recent call last)  
Cell In[35], line 1  
----> 1 complex('10', '20')  
  
TypeError: complex() argument 'real' must be a real number, not str
```

STRING

```
In [36]: str(10)
```

```
Out[36]: '10'
```

```
In [37]: str(10.45)
```

```
Out[37]: '10.45'
```

```
In [ ]: str(
```